

Figure 3: More simple graphs

Figure 3 shows the output graphs from the above code.

Graph varieties

Those aware of graphs will know or at least have heard of a variety of them. Here's a list of some of them, available in Maxima's graphs package (through functions):

- Empty graph (*empty_graph(n)*) – A graph with a given 'n' vertices but no edges.
- Complete graph (*complete_graph(n)*) – A simple graph with all possible edges for a given 'n' vertices.
- Complete bipartite graph (*complete_bipartite_graph(m, n)*) – A simple graph with two sets of vertices, having all possible edges between the vertices from the two sets, but with no edge between the vertices of the same set.
- Cube graph (*cube_graph(n)*) – A graph representing an n-dimensional cube.
- Dodecahedron graph (*dodecahedron_graph()*) – A graph forming a 3-D polyhedron with 12 pentagonal faces.
- Cuboctahedron graph (*cuboctahedron_graph()*) – A graph forming a 3-D polyhedron with eight triangular faces and 12 square faces.
- Icosahedron graph (*icosahedron_graph()*) – A graph forming a 3-D polyhedron with 20 triangular faces.
- Icosidodecahedron graph (*icosidodecahedron_graph()*) – A graph forming a 3-D uniform star polyhedron with 12 star faces and 20 triangular faces.

And here follows a demonstration of the above, along with the visuals (left to right, top to bottom) in Figure 4:

```
$ maxima -q
(%i1) load(graphs)$
...
@ errors, @ warnings
(%i2) g: empty_graph(5);
(%o2)          GRAPH(5 vertices, 0 edges)
(%i3) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i4) g: complete_graph(5);
(%o4)          GRAPH(5 vertices, 10 edges)
(%i5) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i6) g: complete_bipartite_graph(5, 3);
(%o6)          GRAPH(8 vertices, 15 edges)
(%i7) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i8) g: cube_graph(3);
```

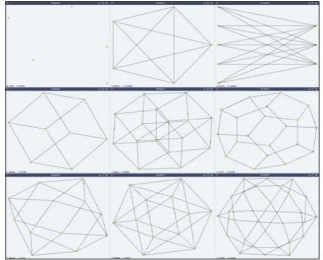


Figure 4: Graph varieties

```
(%o8)          GRAPH(8 vertices, 12 edges)
(%i9) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i10) g: cube_graph(4);
(%o10)          GRAPH(16 vertices, 32 edges)
(%i11) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i12) g: dodecahedron_graph();
(%o12)          GRAPH(20 vertices, 30 edges)
(%i13) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i14) g: cuboctahedron_graph();
(%o14)          GRAPH(12 vertices, 24 edges)
(%i15) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i16) g: icosahedron_graph();
(%o16)          GRAPH(12 vertices, 30 edges)
(%i17) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i18) g: icosidodecahedron_graph();
(%o18)          GRAPH(30 vertices, 60 edges)
(%i19) draw_graph(g, show_id=true, vertex_color=yellow)$
(%i20) quit();
```

Graph beauties

Graphs are really beautiful to visualise. Some of the many beautiful graphs available in Maxima's graphs package (through functions) are listed below:

- Circulant graph (*circulant_graph(n, [x, y, ...])*) – A graph with vertices 0, ..., n-1, where every vertex is adjacent to its x^{th} , y^{th} , ... vertices. Visually, it has a cyclic group of symmetries.
- Flower graph (*flower_snark(n)*) – A graph like a flower with 'n' petals and $4*n$ vertices.
- Wheel graph (*wheel_graph(n)*) – A graph like a wheel with 'n' vertices.
- Clebsch graph (*clebsch_graph()*) – Another symmetrical graph beauty, named by J J Seidel.
- Frucht graph (*frucht_graph()*) – A graph with 12 vertices, 18 edges and no non-trivial symmetries, such that every vertex has three neighbours. It is named after Robert Frucht.